

Chapter 7: Orthogonal Polynomials

Special Functions of Mathematical Physics: A Tourist's Guidebook

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June 12, 2026

Outline

- 1 Spherical Harmonics
- 2 Legendre Polynomial
- 3 Hermite Polynomial
- 4 Laguerre Polynomial
- 5 Chebyshev Polynomial

Where we are going

This chapter studies four families of **orthogonal polynomials**—Legendre, Hermite, Laguerre, and Chebyshev—each the polynomial solution of a second-order differential equation that arises by separating variables in a physical problem. They appear in Laplace, Poisson, Helmholtz, and Schrödinger equations and their time-dependent variants.

- **Legendre** P_n : Laplace's equation $\nabla^2\psi = 0$ in spherical coordinates (spherical harmonics).
- **Hermite** H_n : Schrödinger's equation for the quantum harmonic oscillator.
- **Laguerre** L_n : arises in the radial part of the hydrogen atom and related problems.
- **Chebyshev** T_n : defined through $\cos n\theta$; a natural interval basis, and tied to separated problems in elliptic coordinates.

Across the chapter we use the same toolkit: a defining ODE, a series solution when useful, a *generating function*, a *recurrence*, and an *orthogonality* relation. For Legendre, Hermite, and Laguerre we also record Rodrigues' formulas and Schläfli (contour) integrals.

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Warm-up: 2D Laplace in polar coordinates

A function satisfying Laplace's equation $\nabla^2\psi = 0$ is called a *harmonic function*. In two dimensions the real and imaginary parts of an analytic function are harmonic conjugates, so the polar calculation below is the same separation mechanism that underlies power series in $z = re^{i\theta}$. As a review we solve the two-dimensional Laplace equation in polar coordinates,

$$\nabla^2\psi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial\psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2\psi}{\partial\theta^2} = 0, \quad (7.1.1)$$

by separation of variables. Taking $\psi(r, \theta) = u(r)v(\theta)$ gives

$$v \frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right) + \frac{1}{r^2} u \frac{d^2v}{d\theta^2} = 0.$$

Multiplying by $r^2/(uv)$ and separating,

$$-\frac{v''}{v} = \frac{r(ru')'}{u} = \lambda$$

for an arbitrary constant λ .

Warm-up: 2D Laplace in polar coordinates (cont.)

Solving $v'' + \lambda v = 0$ gives $v(\theta) = e^{\pm i\sqrt{\lambda}\theta}$. Take $\lambda = n^2$ for integer values n to satisfy the periodic boundary conditions in the angular direction. For $u(r)$ the radial equation is

$$r(ru')' - n^2u = 0.$$

Trying $u = r^p$ gives $r(rp r^{p-1})' = p^2 r^p$, so $p^2 = n^2$ and $p = \pm n$. For $n \geq 1$ the radial solutions are r^n and r^{-n} ; for $n = 0$ they are 1 and $\log r$. Inside a disk that includes the origin we keep only the finite powers $r^{|n|}$. On an annulus or exterior region we may also keep negative powers. In the analytic part this is written compactly as

$$\psi(r, \theta) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta} = \sum_{n=0}^{\infty} a_n z^n \quad (7.1.2)$$

for $z = r e^{i\theta}$, since $z^n = r^n e^{in\theta}$, and the coefficients a_n are determined from boundary conditions. A general real harmonic function can also include the conjugate analytic part in powers of $\bar{z}$; annular or exterior regions may also include negative powers.

3D Laplace in spherical coordinates

Now the three-dimensional Laplace equation in spherical coordinates, with $\psi(r, \theta, \phi)$:

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = 0. \quad (7.1.3)$$

Take $\psi(r, \theta, \phi) = u(r) y(\theta, \phi) = u(r) v(\theta) w(\phi)$. Substituting and multiplying by $r^2/(uy)$ separates the radial part:

$$\frac{(r^2 u')'}{u} = - \frac{\frac{1}{\sin \theta} (\sin \theta y_\theta)_\theta + \frac{1}{\sin^2 \theta} y_{\phi\phi}}{y} = \lambda. \quad (7.1.4)$$

For the radial factor we solve $(r^2 u')' - \lambda u = 0$. Taking the ansatz $u = r^n$ we have $r^2 u' = r^2 \cdot n r^{n-1} = n r^{n+1}$, so $(r^2 u')' = n(n+1)r^n$. Substituting gives $n(n+1)r^n = \lambda r^n$, so

$$\lambda = n(n+1).$$

3D Laplace in spherical coordinates (cont.)

The angular factor $y(\theta, \phi)$ then obeys

$$\frac{1}{\sin \theta} (\sin \theta y_\theta)_\theta + \frac{y_{\phi\phi}}{\sin^2 \theta} = -n(n+1) y.$$

Taking $y(\theta, \phi) = v(\theta) w(\phi)$ gives

$$\sin \theta (\sin \theta v')' w + v w'' + n(n+1) \sin^2 \theta v w = 0.$$

Dividing by $y = vw$,

$$\frac{\sin \theta (\sin \theta v')' + n(n+1) \sin^2 \theta v}{v} = -\frac{w''}{w} = \gamma$$

for some γ . The solution to $w'' + \gamma w = 0$ is $w = e^{\pm i\sqrt{\gamma}\phi}$. Periodicity in ϕ forces $\sqrt{\gamma}$ to be a nonnegative integer. Writing the signed Fourier mode as $m \in \mathbb{Z}$, this means $\gamma = m^2$ and $w = e^{im\phi}$.

The associated Legendre equation

Finally, we solve the θ -equation

$$\sin \theta (\sin \theta v')' + n(n+1) \sin^2 \theta v = m^2 v$$

for $v(\theta)$. We simplify with the change of variables $x = \cos \theta$, for which

$$\frac{d}{d\theta} = \frac{d}{dx} \frac{dx}{d\theta} = -\sin \theta \frac{d}{dx}. \quad (7.1.5)$$

From (7.1.5), multiplying by $\sin \theta$ gives the operator identity $\sin \theta \frac{d}{d\theta} = -\sin^2 \theta \frac{d}{dx}$.

Starting with

$$\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{dv}{d\theta} \right) + n(n+1) \sin^2 \theta v = m^2 v$$

we apply this identity once to the inner factor, $\sin \theta \frac{dv}{d\theta} = -\sin^2 \theta \frac{dv}{dx}$, and once more to the outer derivative, obtaining

$$-\sin^2 \theta \frac{d}{dx} \left(-\sin^2 \theta \frac{dv}{dx} \right) + n(n+1) \sin^2 \theta v = m^2 v.$$

Because $\sin^2 \theta = 1 - \cos^2 \theta = 1 - x^2$, this becomes

$$(1 - x^2)((1 - x^2)v')' + n(n+1)(1 - x^2)v - m^2 v = 0.$$

The associated Legendre equation (cont.)

Dividing through by $(1 - x^2)$ puts the equation in standard form:

$$\left((1 - x^2)v'\right)' + \left[n(n+1) - \frac{m^2}{1 - x^2}\right]v = 0. \quad (7.1.6)$$

We call (7.1.6) the *associated Legendre equation*. The angular factor is denoted by

$$v(\theta) = P_n^m(\cos \theta),$$

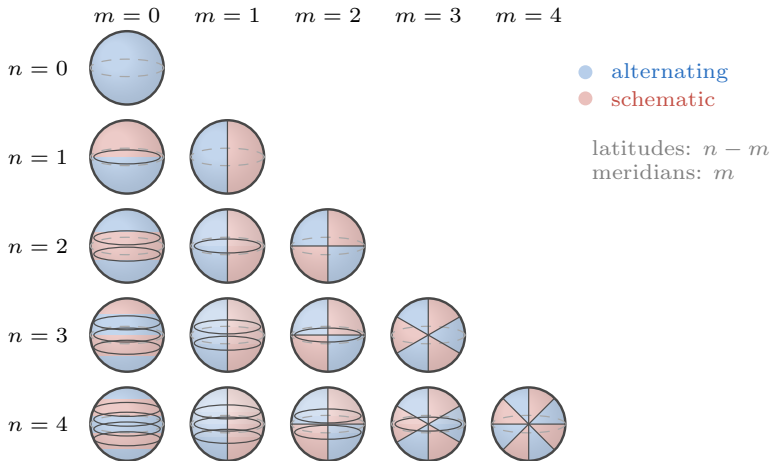
where $P_n^m(x)$ is an *associated Legendre function*. For integer n and $0 \leq |m| \leq n$ it is often grouped with the associated Legendre polynomials used in spherical harmonics. The solution to the three-dimensional Laplace equation is therefore

$$\begin{aligned} \psi(r, \theta, \phi) &= \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm} r^n e^{im\phi} P_n^m(\cos \theta) \\ &= \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm} r^n Y_n^m(\theta, \phi). \end{aligned} \quad (7.1.7)$$

The associated Legendre equation (cont.)

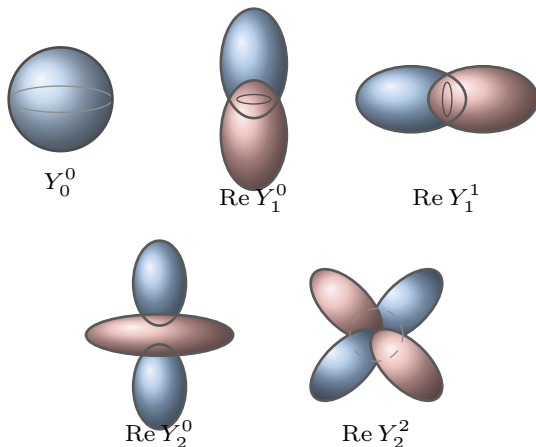
The functions $Y_n^m(\theta, \phi) = e^{im\phi} P_n^m(\cos \theta)$ are *unnormalized* spherical harmonics. Standard normalized spherical harmonics multiply this expression by a constant and use a fixed convention for negative m . This is the regular interior expansion; exterior solutions use the radial power r^{-n-1} instead of r^n .

Low spherical-harmonic nodal patterns



Rows: degree n ; columns: order m . Colors are schematic. Nodal counts: $n - m$ latitude circles and m meridian planes.

Real-part spherical-harmonic surfaces



blue/red show sign;
lobe size shows $|\text{Re } Y_n^m|$

zeros become nodal
latitude/meridian curves

The book's surface picture emphasizes amplitude and sign, not only nodal counts. Blue and red indicate opposite signs of $\text{Re } Y_n^m$; the lobes shrink to zero on the nodal latitude and meridian curves.

Axisymmetry: the ordinary Legendre equation

If the problem is axisymmetric about the polar axis, then nothing depends on ϕ , so $m = 0$. This is weaker than true radial symmetry: true radial symmetry would leave only the $n = 0$ mode. With $m = 0$, the associated Legendre equation (7.1.6) is simply the *Legendre equation*

$$\left((1-x^2)v'\right)' + n(n+1)v = 0, \quad (7.1.8)$$

which has the Legendre polynomial solutions

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{2^n k! (n-2k)! (n-k)!} x^{n-2k}. \quad (7.1.9)$$

The rest of this section develops the Legendre polynomials P_n in detail: their generating function, recurrence, Rodrigues' formula, contour integral, and orthogonality.

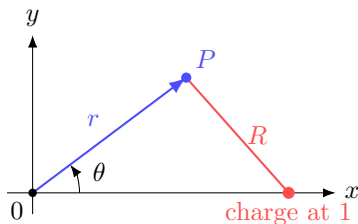
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Generating function from the Coulomb potential

Suppose a point charge sits one unit from the origin along the polar axis. For field points with $|r| < 1$, the electrostatic Coulomb potential $\varphi(r, \theta)$ is proportional to $1/R$, where R is the distance to the point charge. The law of cosines gives

$$\varphi(r, \theta) = \frac{C}{R} = \frac{C}{\sqrt{r^2 + 1 - 2r \cos \theta}}, \quad (7.2.1)$$



$$R^2 = 1 + r^2 - 2r \cos \theta$$

Generating function from the Coulomb potential (cont.)

where C collects the charge magnitude and Coulomb's constant. Inside the source-free ball $r < 1$ it satisfies Laplace's equation, so it expands in Legendre polynomials:

$$\varphi(r, \theta) = \frac{C}{\sqrt{r^2 + 1 - 2r \cos \theta}} = \sum_{n=0}^{\infty} a_n P_n(\cos \theta) r^n.$$

The constant C is a mere scaling factor, so rescale $\{a_n\}$ to set $C = 1$:

$$\frac{1}{\sqrt{r^2 + 1 - 2r \cos \theta}} = \sum_{n=0}^{\infty} a_n P_n(\cos \theta) r^n. \quad (7.2.2)$$

Generating function from the Coulomb potential (cont.)

To fix the a_n , set $\theta = 0$ (so $\cos \theta = 1$):

$$\frac{1}{\sqrt{r^2 + 1 - 2r}} = \sum_{n=0}^{\infty} a_n P_n(1) r^n.$$

We normalize the Legendre polynomials so that $P_n(1) = 1$, giving

$$\frac{1}{\sqrt{r^2 + 1 - 2r}} = \frac{1}{|1 - r|} = \sum_{n=0}^{\infty} a_n r^n.$$

For $0 \leq r < 1$, the left-hand side is $1/(1 - r)$ and expands as the geometric series $1 + r + r^2 + r^3 + \cdots$, so $a_n = 1$ for all n . Hence, for $|r| < 1$,

$$\frac{1}{\sqrt{r^2 + 1 - 2r \cos \theta}} = \sum_{n=0}^{\infty} P_n(\cos \theta) r^n.$$

Taking $t = r$ and $x = \cos \theta$ gives the *generating function* for the Legendre polynomials:

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n. \quad (7.2.3)$$

Three-term recurrence

Differentiate the generating function (7.2.3) with respect to t :

$$\frac{\partial g}{\partial t} = \frac{x - t}{(1 - 2xt + t^2)^{3/2}} = \sum_{n=1}^{\infty} n P_n(x) t^{n-1}. \quad (7.2.4)$$

Equivalently, since $g = (1 - 2xt + t^2)^{-1/2}$,

$$\frac{x - t}{(1 - 2xt + t^2)^{1/2}} = (1 - 2xt + t^2) \sum_{n=1}^{\infty} n P_n(x) t^{n-1},$$

i.e. $(x - t)g(x, t) = (1 - 2xt + t^2) \sum_{n=1}^{\infty} n P_n(x) t^{n-1}$, or

$$(x - t) \sum_{n=0}^{\infty} P_n t^n = (1 - 2xt + t^2) \sum_{n=1}^{\infty} n P_n t^{n-1},$$

writing $P_n \equiv P_n(x)$. Distributing terms into the sums gives

$$\sum_{n=0}^{\infty} x P_n t^n - \sum_{n=0}^{\infty} P_n t^{n+1} = \sum_{n=1}^{\infty} n P_n t^{n-1} - 2 \sum_{n=1}^{\infty} n x P_n t^n + \sum_{n=1}^{\infty} n P_n t^{n+1}. \quad (7.2.5)$$

Three-term recurrence (cont.)

Shift the indices n so the powers t^n line up:

$$\sum_{n=0}^{\infty} x P_n t^n - \sum_{n=0}^{\infty} P_{n-1} t^n = \sum_{n=0}^{\infty} (n+1) P_{n+1} t^n - 2 \sum_{n=0}^{\infty} n x P_n t^n + \sum_{n=0}^{\infty} (n-1) P_{n-1} t^n.$$

It is understood that $P_{-1}(x) = 0$. Combining,

$$\sum_{n=0}^{\infty} [x P_n - P_{n-1} - (n+1) P_{n+1} + 2n x P_n - (n-1) P_{n-1}] t^n = 0.$$

We simplify the expression to get

$$\sum_{n=0}^{\infty} [(2n+1)x P_n(x) - (n+1)P_{n+1}(x) - n P_{n-1}] t^n = 0. \quad (7.2.6)$$

Because this holds for all t , each coefficient vanishes:

$(2n+1)x P_n - (n+1)P_{n+1} - n P_{n-1} = 0$. Hence the *three-term recurrence*

$$(n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x). \quad (7.2.7)$$

First few Legendre polynomials

Starting from $P_0(x) = 1$ and $P_1(x) = x$, the recurrence (7.2.7) generates the rest:

$$P_0(x) = 1,$$

$$P_1(x) = x,$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1),$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

For example, with $n = 1$ in (7.2.7),

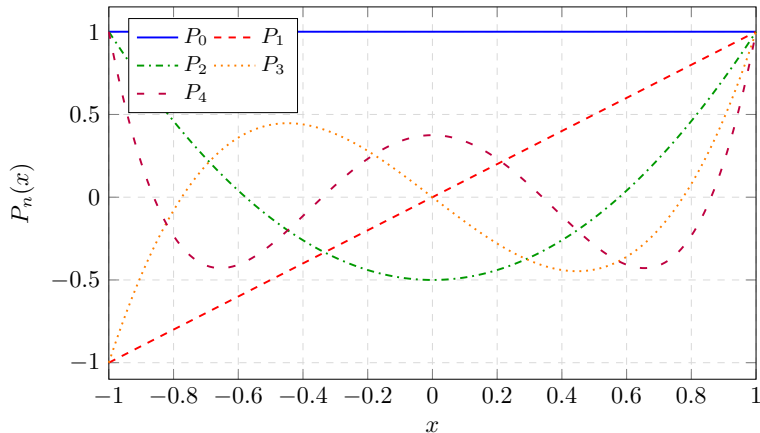
$$2 P_2 = 3x P_1 - 1 \cdot P_0 = 3x^2 - 1 \implies P_2 = \frac{1}{2}(3x^2 - 1),$$

and with $n = 2$,

$$3 P_3 = 5x P_2 - 2 P_1 = \frac{5x}{2}(3x^2 - 1) - 2x = \frac{1}{2}(15x^3 - 5x - 4x) = \frac{1}{2}(15x^3 - 9x),$$

$$\text{so } P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

Shapes of the first Legendre polynomials



Each P_n is bounded on $[-1, 1]$ and normalized by $P_n(1) = 1$. Increasing n adds interior zeros while preserving the Sturm–Liouville orthogonality on the same interval.

Contour (Schl fli) integral

Divide the generating function (7.2.3) by t^{m+1} and integrate around a contour γ about the origin in the complex plane:

$$\oint_{\gamma} \frac{1}{t^{m+1} \sqrt{1 - 2xt + t^2}} dt = \oint_{\gamma} \sum_{n=0}^{\infty} P_n(x) t^{n-(m+1)} dt = 2\pi i P_m(x),$$

since only the $n = m$ term contributes a residue (the coefficient of t^{-1}). This gives the contour-integral representation

$$P_m(x) = \frac{1}{2\pi i} \oint_{\gamma} \frac{1}{t^{m+1} \sqrt{1 - 2xt + t^2}} dt. \quad (7.2.8)$$

Contour (Schl fli) integral (cont.)

Choosing the contour. The function $\sqrt{1 - 2xt + t^2}$ has branch points where $1 - 2xt + t^2 = 0$, i.e. at $t = x \pm i\sqrt{1 - x^2}$. Taking $x = \cos \theta$, the branch points are at $t = \cos \theta \pm i \sin \theta = e^{\pm i\theta}$, so they lie on the *unit circle*. We take one branch cut joining the two branch points along an arc of the unit circle, not through the origin, and choose γ to be any small circle around the origin inside the unit circle, so it avoids the branch points and the cut.

Rodrigues' formula

The binomial expansion of $(1 - x^2)^n$ is

$$(1 - x^2)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} x^{2k} = \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)! k!} x^{2k}. \quad (7.2.9)$$

Differentiate n times. Each term x^{2k} contributes only if $2k \geq n$, since

$\frac{d^n}{dx^n} x^{2k} = \frac{(2k)!}{(2k-n)!} x^{2k-n}$ vanishes once $2k < n$. The surviving terms have

$\lceil n/2 \rceil \leq k \leq n$; relabelling the sum by $k \mapsto n - k$ sends this range to $0 \leq k \leq \lfloor n/2 \rfloor$ and turns the exponent $x^{2(n-k)-n} = x^{n-2k}$, the factor $\binom{n}{k} \rightarrow \binom{n}{n-k} = \binom{n}{k}$ with $k! \rightarrow (n-k)!$ and $(n-k)! \rightarrow k!$, and $(2k)!/(2k-n)! \rightarrow (2n-2k)!/(n-2k)!$, giving

$$\begin{aligned} \frac{d^n}{dx^n} (1 - x^2)^n &= \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^{n-k} \frac{n!}{(n-k)! k!} \frac{(2n-2k)!}{(n-2k)!} x^{n-2k} \\ &= (-1)^n n! \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{k! (n-k)! (n-2k)!} x^{n-2k}. \end{aligned} \quad (7.2.10)$$

Rodrigues' formula (cont.)

The series representation of the Legendre polynomial is

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{2^n k! (n-2k)! (n-k)!} x^{n-2k}.$$

Comparing with (7.2.10), the two differ by the factor $(-1)^n 2^n n!$, and $(-1)^n (1-x^2)^n = (x^2-1)^n$. Therefore

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n, \quad (7.2.11)$$

which we call *Rodrigues' formula*. Recall Cauchy's integral formula for the n th derivative,

$$f^{(n)}(x) = \frac{n!}{2\pi i} \oint \frac{f(t)}{(t-x)^{n+1}} dt.$$

Hence the Schläfli representation

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n = \frac{1}{2^{n+1} \pi i} \oint \frac{(t^2-1)^n}{(t-x)^{n+1}} dt. \quad (7.2.12)$$

Rodrigues' formula (cont.)

The contour is positively oriented and encloses $t = x$. Here we use these formulas only for integer n ; noninteger Legendre functions require a separate branch convention and are not needed below. In that broader setting, the Schläfli representation is one standard starting point for extending the Legendre family beyond polynomial degree; Rodrigues' formula can also be read formally through fractional-calculus derivatives once the same branch choices are fixed.

Orthogonality of Legendre polynomials

The Sturm–Liouville form of the Legendre equation is

$$\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} P_n(x) \right] + n(n+1) P_n(x) = 0. \quad (7.2.13)$$

On the domain $x \in [-1, 1]$ with weight $w = 1$, the inner product

$$(P_n, P_m) = \int_{-1}^1 P_n(x) P_m(x) dx = 0 \quad \text{for } n \neq m.$$

To find the normalization constant we use Rodrigues' formula.

Theorem. If P_m and P_n are the m th and n th degree Legendre polynomials, then

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2m+1} \delta_{nm}.$$

Orthogonality of Legendre polynomials (cont.)

Proof. Without loss of generality suppose $n \leq m$. By Rodrigues' formula,

$$\int_{-1}^1 P_m P_n dx = \frac{1}{2^{m+n} m! n!} \int_{-1}^1 \frac{d^m}{dx^m} (x^2 - 1)^m \frac{d^n}{dx^n} (x^2 - 1)^n dx.$$

Because $(x^2 - 1)^m$ has m th order zeros at $x = \pm 1$, the boundary terms from integration by parts vanish. After integrating by parts once,

$$\int_{-1}^1 P_m P_n dx = \frac{-1}{2^{m+n} m! n!} \int_{-1}^1 \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^m \frac{d^{n+1}}{dx^{n+1}} (x^2 - 1)^n dx.$$

Continuing, the boundary terms again vanish because each derivative of $(x^2 - 1)^m$ that appears at the endpoints still carries enough powers of $(x - 1)$ and $(x + 1)$ to be zero. After integrating by parts a total of n times,

$$\int_{-1}^1 P_m P_n dx = \frac{(-1)^n}{2^{m+n} m! n!} \int_{-1}^1 \frac{d^{m-n}}{dx^{m-n}} (x^2 - 1)^m \frac{d^{2n}}{dx^{2n}} (x^2 - 1)^n dx. \quad (7.2.14)$$

Orthogonality of Legendre polynomials (cont.)

Case $m = n$. Here $\frac{d^{2m}}{dx^{2m}}(x^2 - 1)^m = \frac{d^{2m}}{dx^{2m}}(x^{2m} + \dots) = (2m)!$ (only the top power x^{2m} survives $2m$ derivatives, and $\frac{d^{2m}}{dx^{2m}}x^{2m} = (2m)!$), so (7.2.14) becomes

$$\begin{aligned}\int_{-1}^1 (P_m)^2 dx &= \frac{(-1)^m}{2^{2m}(m!)^2} (2m)! \int_{-1}^1 (x^2 - 1)^m dx \\&= \frac{(-1)^m}{2^{2m}(m!)^2} (2m)! \int_{-1}^1 (x+1)^m (x-1)^m dx \\&= \frac{(-1)^{2m}}{2^{2m}(m!)^2} (2m)! \frac{(m!)^2}{(2m)!} \int_{-1}^1 (x-1)^{2m} dx \\&= \frac{1}{2^{2m}} \cdot \frac{2^{2m+1}}{2m+1} = \frac{2}{2m+1}.\end{aligned}$$

The third line integrates $\int (x+1)^m (x-1)^m dx$ by parts m more times, each time differentiating the $(x+1)$ factor and antidifferentiating the $(x-1)$ factor; the boundary terms vanish because $(x+1)^m$ and its derivatives keep a zero at $x = -1$ while the antiderivatives of $(x-1)^m$ keep a zero at $x = +1$. The m sign flips contribute $(-1)^m$ (turning the prefactor $(-1)^m$ into $(-1)^{2m} = 1$); fully differentiating gives

Orthogonality of Legendre polynomials (cont.)

$\frac{d^m}{dx^m}(x+1)^m = m!$; and antidifferentiating $(x-1)^m$ a total of m times gives $\frac{m!}{(2m)!}(x-1)^{2m}$, the source of the factor $(m!)^2/(2m)!$. The last line uses

$$\int_{-1}^1 (x-1)^{2m} dx = \left[\frac{(x-1)^{2m+1}}{2m+1} \right]_{-1}^1 = \frac{2^{2m+1}}{2m+1}.$$

Orthogonality of Legendre polynomials (cont.)

Case $m \neq n$. It follows that $n < m$ (i.e. $m > n$), in which case we can integrate (7.2.14) by parts at least once more. Doing so,

$$\int_{-1}^1 P_m P_n dx = \frac{(-1)^{n+1}}{2^{m+n} m! n!} \int_{-1}^1 \frac{d^{m-n-1}}{dx^{m-n-1}} (x^2 - 1)^m \frac{d^{2n+1}}{dx^{2n+1}} (x^2 - 1)^n dx.$$

The term $\frac{d^{2n+1}}{dx^{2n+1}} (x^2 - 1)^n = 0$, because $(x^2 - 1)^n$ is a polynomial of degree $2n$, annihilated by a derivative of order greater than $2n$. So the whole integral is zero. Therefore

$$(P_n, P_m) = \frac{2}{2m+1} \delta_{nm}. \quad \square$$

Fourier–Legendre series

Recall the harmonic oscillator equation $u'' + n^2 u = 0$, with solutions $u_n = e^{inx}$ orthogonal under $(u, v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u \bar{v} dx$, giving the Fourier series $f(x) = \sum_{n=-\infty}^{\infty} (f, e^{inx}) e^{inx}$.

The Legendre polynomials $\{P_n\}$ are orthogonal, $(P_n, P_m) = \frac{2}{2m+1} \delta_{nm}$, and *complete*: any $f \in L^2[-1, 1]$ expands in the L^2 sense as

$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x).$$

To find $\{a_n\}$, take the inner product with P_m :

$$(f, P_m) = \left(\sum_{n=0}^{\infty} a_n P_n, P_m \right) = \sum_{n=0}^{\infty} a_n \frac{2}{2m+1} \delta_{nm} = a_m \frac{2}{2m+1}.$$

Fourier–Legendre series (cont.)

The coefficient

$$a_m = \frac{2m+1}{2} (f, P_m)$$

is the projection of f onto P_m . From this we have the *Fourier–Legendre series*

$$f(x) = \sum_{n=0}^{\infty} \frac{2n+1}{2} (f, P_n) P_n(x). \quad (7.2.15)$$

Useful formulas for the Legendre polynomial

differential equation $(1 - x^2)P_m'' - 2xP_m' + m(m+1)P_m = 0$

series solution
$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2n-2k)!}{2^n k! (n-2k)! (n-k)!} x^{n-2k}$$

generating function
$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

recurrence $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$

orthogonality
$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2m+1} \delta_{nm}$$

Rodrigues
$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Schläfli
$$P_n(x) = \frac{1}{2^{n+1} \pi i} \oint \frac{(t^2 - 1)^n}{(t - x)^{n+1}} dt$$

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The Hermite equation and series solution

We look at polynomial solutions of the *Hermite differential equation*

$$\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2n y = 0. \quad (7.3.1)$$

Put $y = \sum_{j=0}^{\infty} a_j x^j$. Then

$$y' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j, \quad y'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j.$$

The term $-2xy'$ contributes $-2ja_j x^j$ for $j \geq 1$ and contributes 0 for $j = 0$, so the coefficient of x^j in (7.3.1) is

$$(j+2)(j+1)a_{j+2} + 2(n-j)a_j = 0.$$

Hence

$$a_{j+2} = \frac{2(j-n)}{(j+2)(j+1)} a_j.$$

The Hermite equation and series solution (cont.)

Even coefficients depend only on a_0 , and odd coefficients depend only on a_1 . The physically useful polynomial case occurs when n is a nonnegative integer, because one chain terminates at $j = n$. With the conventional leading coefficient 2^n , this gives

$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}, \quad (7.3.2)$$

where $\lfloor n/2 \rfloor = n/2$ if n is even and $(n-1)/2$ if n is odd.

The Hermite equation and series solution (cont.)

Multiplying (7.3.1) by the integrating factor e^{-x^2} puts it in Sturm–Liouville form,

$$\frac{d}{dx} \left(e^{-x^2} \frac{dy}{dx} \right) + 2n e^{-x^2} y = 0. \quad (7.3.3)$$

With weight $w(x) = e^{-x^2}$ and domain $x \in (-\infty, \infty)$, the eigenfunctions are orthogonal under $(u, v)_w = \int_{-\infty}^{\infty} uv e^{-x^2} dx$:

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = 0 \quad \text{if } n \neq m. \quad (7.3.4)$$

The norm comes from the generating function. Multiplying two copies and integrating,

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-t^2+2tx} e^{-s^2+2sx} dx = e^{-t^2-s^2} \int_{-\infty}^{\infty} e^{-(x-t-s)^2+(t+s)^2} dx = \sqrt{\pi} e^{2st}.$$

Comparing coefficients of $s^m t^n$ gives

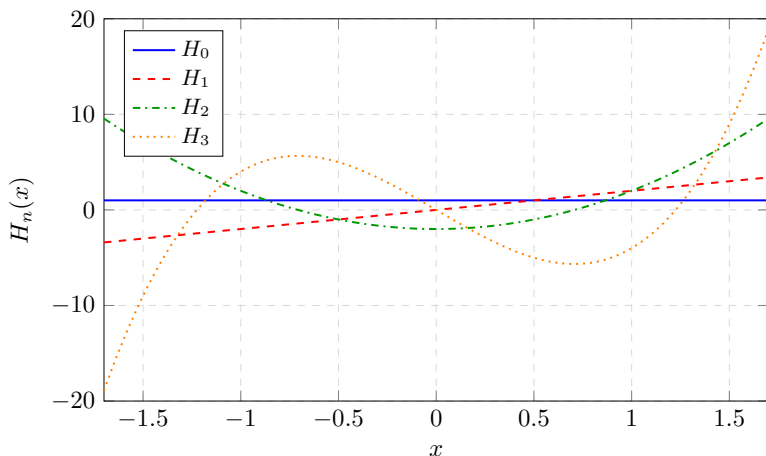
$$\int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx = 2^n n! \sqrt{\pi} \delta_{mn}.$$

The Hermite equation and series solution (cont.)

Defining $\psi_n(x) = e^{-x^2/2} H_n$ gives functions orthogonal under the ordinary Euclidean inner product:

$$\int_{-\infty}^{\infty} \psi_m(x) \psi_n(x) dx = \int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx = 0 \quad (m \neq n).$$

Shapes of the first Hermite polynomials



The Hermite polynomials grow rapidly away from the origin. In the harmonic oscillator the physical functions are $e^{-x^2/2}H_n(x)$, where the Gaussian factor restores decay.

Which equation does $\psi_n = e^{-x^2/2} H_n$ solve?

Take $\psi = e^{-x^2/2} H_n = e^{-x^2/2} y$, where $y = H_n$ satisfies $y'' - 2xy' + 2ny = 0$. Invert to $y = e^{x^2/2} \psi$; then (using $\frac{d}{dx} e^{x^2/2} = x e^{x^2/2}$)

$$y' = e^{x^2/2} (\psi' + x\psi),$$

$$y'' = e^{x^2/2} (\psi'' + 2x\psi' + (x^2 + 1)\psi).$$

Substitute into $y'' - 2xy' + 2ny = 0$ and divide by $e^{x^2/2}$:

$$(\psi'' + 2x\psi' + (x^2 + 1)\psi) - 2x(\psi' + x\psi) + 2n\psi = 0.$$

The $2x\psi'$ terms cancel and $x^2 + 1 - 2x^2 = 1 - x^2$, leaving the quantum harmonic oscillator equation

$$\frac{d^2}{dx^2} \psi_n + (2n + 1 - x^2) \psi_n = 0. \quad (7.3.5)$$

The solutions are $\psi_n(x) = e^{-x^2/2} H_n(x)$.

Example: the quantum harmonic oscillator

The one-dimensional Schrödinger equation for a particle in a potential is

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \Psi(x, t), \quad (7.3.6)$$

where $V(x)$ is the potential, m the particle mass, and $\hbar$ the reduced Planck constant. Separating $\Psi(x, t) = u(t)\psi(x)$,

$$\frac{i\hbar u'}{u} = \frac{-\frac{\hbar^2}{2m}\psi'' + V(x)\psi}{\psi} = E,$$

where E is the energy of a state. From $i\hbar u' = Eu$ we have

$$u(t) = a e^{-iEt/\hbar}.$$

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi.$$

Example: the quantum harmonic oscillator (cont.)

The operator $H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$ is the *Hamiltonian*. A harmonic oscillator has $V(x) = \frac{1}{2} kx^2$ (Hooke's law), and with $\omega = \sqrt{k/m}$ this is $V(x) = \frac{1}{2} m\omega^2 x^2$. The Schrödinger equation becomes

$$-\frac{\hbar^2}{2m} \psi'' + \frac{1}{2} m\omega^2 x^2 \psi = E\psi,$$

which we rewrite as

$$\psi'' + \left(\frac{2mE}{\hbar^2} - \frac{m^2\omega^2}{\hbar^2} x^2 \right) \psi = 0.$$

Making the change of variable $\tilde{x} = \sqrt{m\omega/\hbar} x$, so that $\frac{d}{dx} = \sqrt{m\omega/\hbar} \frac{d}{d\tilde{x}}$ and hence

$\frac{d^2}{dx^2} = \frac{m\omega}{\hbar} \frac{d^2}{d\tilde{x}^2}$, and noting $\frac{m^2\omega^2}{\hbar^2} x^2 = \frac{m\omega}{\hbar} \tilde{x}^2$, divide through by $\frac{m\omega}{\hbar}$:

$$\frac{d^2\psi}{d\tilde{x}^2} + \left(\frac{2E}{\hbar\omega} - \tilde{x}^2 \right) \psi = 0,$$

since $\frac{2mE}{\hbar^2} \cdot \frac{\hbar}{m\omega} = \frac{2E}{\hbar\omega}$.

Example: the quantum harmonic oscillator (cont.)

Comparing with (7.3.5), we read off

$$\psi_n(\tilde{x}) = e^{-\tilde{x}^2/2} H_n(\tilde{x}), \quad \frac{2E}{\hbar\omega} = 2n + 1.$$

Physically meaningful (non-divergent) solutions require n to be a nonnegative integer, giving the quantized energies

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega. \quad (7.3.7)$$

The full solution is

$$\Psi(x, t) = \sum_{n=0}^{\infty} a_n e^{-i\omega(n+\frac{1}{2})t} e^{-\frac{1}{2}m\omega\hbar^{-1}x^2} H_n\left(\sqrt{m\omega\hbar^{-1}} x\right). \quad (7.3.8)$$

Generating function, Rodrigues, Schlöfli

The generating function for the Hermite polynomials is

$$g(x, t) = e^{-(t-x)^2} e^{x^2} = e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}. \quad (7.3.9)$$

Differentiate m times with respect to t :

$$e^{x^2} \frac{d^m}{dt^m} e^{-(t-x)^2} = H_m(x) + H_{m+1}(x)t + \frac{1}{2}H_{m+2}(x)t^2 + \dots$$

Setting $t = 0$ kills every term on the right except the constant one, leaving

$e^{x^2} \frac{d^m}{dt^m} e^{-(t-x)^2} \Big|_{t=0} = H_m(x)$. Now change variable to $s = x - t$, so that $\partial s / \partial t = -1$ and hence $d/dt = -d/ds$, giving $d^m/dt^m = (-1)^m d^m/ds^m$; at $t = 0$ we have $s = x$. Thus

$$e^{x^2} \frac{d^m}{dt^m} e^{-(x-t)^2} \Big|_{t=0} = (-1)^m e^{x^2} \frac{d^m}{ds^m} e^{-s^2} \Big|_{s=x} = (-1)^m e^{x^2} \frac{d^m}{dx^m} e^{-x^2} = H_m(x),$$

giving *Rodrigues' formula*

$$H_m(x) = (-1)^m e^{x^2} \frac{d^m}{dx^m} e^{-x^2}. \quad (7.3.10)$$

Generating function, Rodrigues, Schläfli (cont.)

Divide the generating function by t^{m+1} and integrate along γ about the origin:

$$\oint_{\gamma} \frac{e^{-t^2+2tx}}{t^{m+1}} dt = \oint_{\gamma} \sum_{n=0}^{\infty} H_n(x) \frac{t^{n-m-1}}{n!} dt = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} 2\pi i \delta_{nm} = \frac{H_m(x)}{m!} 2\pi i.$$

Hence the *Schläfli integral*

$$H_m(x) = \frac{m!}{2\pi i} \oint_{\gamma} \frac{e^{-t^2+2tx}}{t^{m+1}} dt. \quad (7.3.11)$$

Example: special values $H_n(0)$

From the generating function at $x = 0$,

$$g(0, t) = e^{-t^2} = \sum_{n=0}^{\infty} H_n(0) \frac{t^n}{n!}.$$

The Taylor series of e^{-t^2} also gives

$$e^{-t^2} = \sum_{m=0}^{\infty} (-1)^m \frac{t^{2m}}{m!}.$$

Matching coefficients of t^{2m} : the odd powers have no counterpart, while $\frac{H_{2m}(0)}{(2m)!} = \frac{(-1)^m}{m!}$. So

$$H_{2n+1}(0) = 0 \quad \text{and} \quad H_{2n}(0) = (-1)^n \frac{(2n)!}{n!}. \quad (7.3.12)$$

Recurrence relations

Differentiate the generating function with respect to x in two ways. Term by term,

$$\frac{\partial g}{\partial x} = \sum_{n=0}^{\infty} H'_n \frac{t^n}{n!}, \text{ while from } g = e^{-t^2+2tx},$$

$$\frac{\partial g}{\partial x} = 2t e^{-t^2+2tx} = 2t \sum_{n=0}^{\infty} H_n \frac{t^n}{n!} = 2 \sum_{n=0}^{\infty} H_n \frac{t^{n+1}}{n!}.$$

Equating the two expressions,

$$\sum_{n=0}^{\infty} H'_n \frac{t^n}{n!} = 2 \sum_{n=0}^{\infty} H_n \frac{t^{n+1}}{n!}.$$

On the right shift $n \mapsto n-1$ so the power becomes t^n , using $\frac{1}{(n-1)!} = \frac{n}{n!}$:

$$\sum_{n=0}^{\infty} H'_n \frac{t^n}{n!} = 2 \sum_{n=1}^{\infty} H_{n-1} \frac{n t^n}{n!}.$$

Matching the coefficient of $t^n/n!$ gives

$$H'_n(x) = 2n H_{n-1}(x). \quad (7.3.13)$$

Recurrence relations (cont.)

Differentiate the generating function with respect to t :

$$\frac{\partial g}{\partial t} = (-2t + 2x) e^{-t^2+2tx} = \sum_{n=1}^{\infty} H_n \frac{t^{n-1}}{(n-1)!}.$$

Thus $(-2t + 2x) \sum_{n=0}^{\infty} H_n \frac{t^n}{n!} = \sum_{n=1}^{\infty} H_n \frac{t^{n-1}}{(n-1)!}$. On the right, shift $n \mapsto n+1$ so the power becomes t^n ; on the left, expand the product:

$$-2 \sum_{n=0}^{\infty} H_n \frac{t^{n+1}}{n!} + 2 \sum_{n=0}^{\infty} x H_n \frac{t^n}{n!} = \sum_{n=0}^{\infty} H_{n+1} \frac{t^n}{n!}.$$

Shift $n \mapsto n-1$ in the first sum (using $\frac{1}{(n-1)!} = \frac{n}{n!}$) so all powers of t agree:

$$-2 \sum_{n=1}^{\infty} H_{n-1} \frac{n t^n}{n!} + 2 \sum_{n=0}^{\infty} x H_n \frac{t^n}{n!} = \sum_{n=0}^{\infty} H_{n+1} \frac{t^n}{n!}.$$

For $n \geq 1$ the coefficient of $t^n/n!$ gives $-2n H_{n-1} + 2x H_n = H_{n+1}$, i.e. the *recurrence*

$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x). \quad (7.3.14)$$

Useful formulas for the Hermite polynomial

differential equation $H_n'' - 2x H_n' + 2n H_n = 0$

series solution
$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}$$

generating function
$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

recurrence
$$H_n' = 2n H_{n-1}$$

orthogonality
$$H_{n+1}(x) = 2x H_n(x) - 2n H_{n-1}(x)$$
$$\int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx = n! 2^n \sqrt{\pi} \delta_{mn}$$

Rodrigues
$$H_m(x) = (-1)^m e^{x^2} \frac{d^m}{dx^m} e^{-x^2}$$

Schl\"afli
$$H_m(x) = \frac{m!}{2\pi i} \oint \frac{e^{-t^2+2tx}}{t^{m+1}} dt$$

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The Laguerre equation and series solution

Solutions of the *Laguerre equation*

$$x y'' + (1 - x) y' + n y = 0 \quad (7.4.1)$$

can be found by a power series. Put $y = \sum_{k=0}^{\infty} a_k x^k$. The coefficient of x^k receives three contributions:

$$x y'' \rightsquigarrow k(k+1)a_{k+1}, \quad y' \rightsquigarrow (k+1)a_{k+1}, \quad -x y' + n y \rightsquigarrow (n-k)a_k.$$

Thus

$$(k+1)^2 a_{k+1} + (n-k)a_k = 0, \quad a_{k+1} = -\frac{n-k}{(k+1)^2} a_k.$$

Taking $a_0 = 1$ gives $a_k = (-1)^k n! / ((n-k)!(k!)^2)$. For a nonnegative integer n , the recurrence stops at $k = n$ and the regular series terminates as the polynomial

$$L_n(x) = \sum_{k=0}^n \frac{(-1)^k n! x^k}{(n-k)!(k!)^2}. \quad (7.4.2)$$

For non-integer parameters the regular solution is no longer this finite polynomial. The generalized (associated) Laguerre polynomials $L_n^{(\alpha)}(x)$ solve the similar equation $x y'' + (\alpha + 1 - x) y' + n y = 0$ for arbitrary real α .

The Laguerre equation and series solution (cont.)

Multiply the Laguerre equation by the integrating factor e^{-x} . Since

$\frac{d}{dx}(xe^{-x}) = (1-x)e^{-x}$, we have $(xe^{-x}y')' = xe^{-x}y'' + (1-x)e^{-x}y'$, so the equation takes the Sturm–Liouville form

$$[xe^{-x}y']' + ne^{-x}y = 0. \quad (7.4.3)$$

The orthogonality is the same Sturm–Liouville subtraction used in Chapter 5. Let $y_m = L_m$ and $y_n = L_n$. Multiply the n equation by y_m , the m equation by y_n , subtract, and integrate:

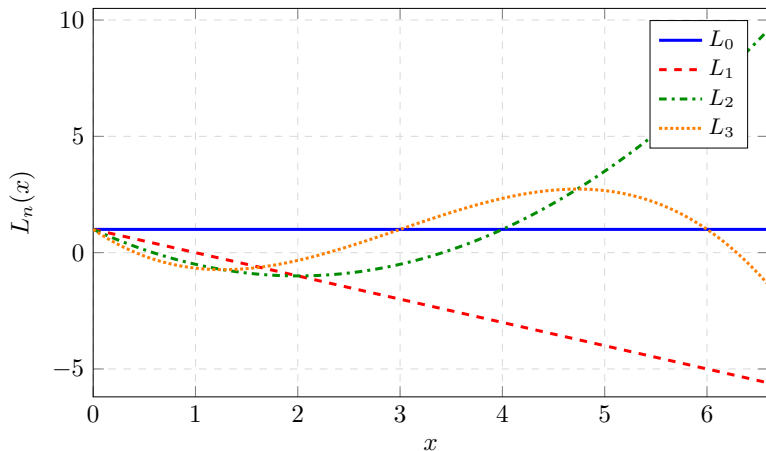
$$(n-m) \int_0^\infty y_m y_n e^{-x} dx = - \int_0^\infty [y_m(xe^{-x}y_n')' - y_n(xe^{-x}y_m')'] dx.$$

The right-hand side is the boundary term $-[xe^{-x}(y_m y_n' - y_n y_m')]_0^\infty$, which is zero because $xe^{-x} = 0$ at $x = 0$ and decays exponentially at infinity while y_m, y_n are polynomials. Therefore,

$$\int_0^\infty L_m(x)L_n(x)e^{-x} dx = 0 \quad \text{if } m \neq n. \quad (7.4.4)$$

The norm is 1 for this standard normalization; we verify it from the generating function on the next frame.

Shapes of the first Laguerre polynomials



All standard Laguerre polynomials satisfy $L_n(0) = 1$. Their positive zeros and the weight e^{-x} are what make them natural on the half-line $[0, \infty)$.

Generating function and Schläfli integral

The generating function for the Laguerre polynomials is

$$g(x, t) = \frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} L_n(x) t^n, \quad |t| < 1. \quad (7.4.5)$$

Note that at $x = 0$,

$$g(0, t) = \frac{1}{1-t} = \sum_{n=0}^{\infty} t^n,$$

so $L_n(0) = 1$. Divide the generating function by t^{m+1} (write z for the contour variable) and integrate about the origin, avoiding the essential singularity at $t = 1$:

$$\oint \frac{g(x, z)}{z^{m+1}} dz = \oint \sum_{n=0}^{\infty} L_n(x) z^{n-m-1} dz = 2\pi i L_m(x).$$

Generating function and Schläfli integral (cont.)

Hence the *Schläfli integral*

$$L_m(x) = \frac{1}{2\pi i} \oint_{|z|=1/2} \frac{e^{-xz/(1-z)}}{(1-z)z^{m+1}} dz. \quad (7.4.6)$$

The same generating function gives the missing norm. Since

$$1 + \frac{s}{1-s} + \frac{t}{1-t} = \frac{1-st}{(1-s)(1-t)},$$

we get

$$\sum_{m,n \geq 0} \left(\int_0^\infty L_m L_n e^{-x} dx \right) s^m t^n = \int_0^\infty e^{-x} g(x, s) g(x, t) dx = \frac{1}{1-st}.$$

Thus $\int_0^\infty L_m L_n e^{-x} dx = \delta_{mn}$.

Generating function and Schläfli integral (cont.)

Take $z = (s - x)/s$ and apply the Cauchy integral formula to obtain *Rodrigues' formula*. Here

$$1 - z = \frac{x}{s}, \quad z^{m+1} = \frac{(s - x)^{m+1}}{s^{m+1}}, \quad dz = \frac{x}{s^2} ds,$$

and the exponential becomes $e^{-xz/(1-z)} = e^{-(s-x)} = e^x e^{-s}$. Therefore

$$\frac{e^{-xz/(1-z)}}{(1-z)z^{m+1}} dz = e^x \frac{s^m e^{-s}}{(s-x)^{m+1}} ds.$$

$$L_m(x) = \frac{e^x}{2\pi i} \oint \frac{s^m e^{-s}}{(s-x)^{m+1}} ds = \frac{e^x}{m!} \frac{d^m}{dx^m} (x^m e^{-x}). \quad (7.4.7)$$

Recurrence from the generating function

Differentiate the generating function (7.4.5) with respect to x . Since $\partial_x g = -\frac{t}{1-t} g$,

$$-\frac{t}{1-t} \sum_{n=0}^{\infty} L_n(x) t^n = \sum_{n=0}^{\infty} L'_n(x) t^n.$$

Multiply through by $(1-t)$:

$$-t \sum_{n=0}^{\infty} L_n(x) t^n = (1-t) \sum_{n=0}^{\infty} L'_n(x) t^n,$$

that is

$$\sum_{n=0}^{\infty} L_n(x) t^{n+1} = \sum_{n=0}^{\infty} L'_n(x) t^{n+1} - \sum_{n=0}^{\infty} L'_n(x) t^n.$$

Recurrence from the generating function (cont.)

Shift the last sum's index $n \mapsto n+1$ (its $n=-1$ term is $L'_0 = 0$ since $L_0 = 1$) so every term carries t^{n+1} , and combine into one sum:

$$\sum_{n=0}^{\infty} (L_n - L'_n + L'_{n+1}) t^{n+1} = 0,$$

so each coefficient vanishes: $L'_{n+1} = L'_n - L_n$, equivalently $L'_n = L'_{n-1} - L_{n-1}$.

A second relation expresses xL'_n in closed form. Reading $x \frac{d}{dx} x^k = k x^k$ off the series

$$L_n(x) = \sum_{k=0}^n \frac{(-1)^k n! x^k}{(n-k)! (k!)^2},$$

$$x L'_n(x) = \sum_{k=0}^n \frac{(-1)^k n! k x^k}{(n-k)! (k!)^2}.$$

Expanding $n L_n - n L_{n-1}$ from the same series and matching the coefficient of x^k (using $n \binom{n}{k} - n \binom{n-1}{k} = k \binom{n}{k}$, the absorption identity) reproduces exactly this sum, so

$$x L'_n(x) = n L_n(x) - n L_{n-1}(x). \quad (7.4.8)$$

This expresses xL'_n in terms of L_n and L_{n-1} .

Recurrence from the generating function (cont.)

Combining these two derivative identities gives the standard three-term recurrence. Use (7.4.8) with $n + 1$:

$$x L'_{n+1} = (n+1)(L_{n+1} - L_n).$$

From $L'_{n+1} = L'_n - L_n$, multiply by x and substitute $xL'_n = n(L_n - L_{n-1})$:

$$(n+1)(L_{n+1} - L_n) = n(L_n - L_{n-1}) - xL_n.$$

Solving for L_{n+1} ,

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x). \quad (7.4.9)$$

Useful formulas for the Laguerre polynomial

differential equation $x L_n'' + (1-x)L_n' + n L_n = 0$

series solution
$$L_n(x) = \sum_{k=0}^n \frac{(-1)^k n! x^k}{(n-k)! (k!)^2}$$

generating function
$$\frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} L_n(x) t^n$$

recurrence $x L_n' = n L_n - n L_{n-1}$

$$(n+1)L_{n+1} = (2n+1-x)L_n - n L_{n-1}$$

orthogonality
$$\int_0^{\infty} L_m L_n e^{-x} dx = \delta_{mn} \quad \text{for this standard normalization}$$

Rodrigues
$$L_m(x) = \frac{e^x}{m!} \frac{d^m}{dx^m} (x^m e^{-x})$$

Schl\"afli
$$L_m(x) = \frac{1}{2\pi i} \oint_{|z|=1/2} \frac{e^{-xz/(1-z)}}{(1-z) z^{m+1}} dz$$

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Definition through $\cos n\theta$

Consider the polynomials formed by taking

$$T_n(\cos \theta) = \cos n\theta. \quad (7.5.1)$$

That is, set $x = \cos \theta$ and read off a polynomial in x . The first cases:

$$T_0(x) = \cos 0 = 1, \quad T_1(x) = \cos \theta = x.$$

For $\cos 2\theta$,

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 2x^2 - 1,$$

so $T_2(x) = 2x^2 - 1$.

Definition through $\cos n\theta$ (cont.)

For $\cos 3\theta$, use $\cos 3\theta + i \sin 3\theta = e^{i3\theta} = (\cos \theta + i \sin \theta)^3$. Taking the real part,

$$\begin{aligned} T_3(\cos \theta) &= \operatorname{Re}(\cos \theta + i \sin \theta)^3 \\ &= \operatorname{Re}(\cos^3 \theta + 3i \sin \theta \cos^2 \theta - 3 \sin^2 \theta \cos \theta - i \sin^3 \theta) \\ &= \cos^3 \theta - 3(1 - \cos^2 \theta) \cos \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta. \end{aligned}$$

Therefore

$$T_3(x) = 4x^3 - 3x. \tag{7.5.2}$$

The name has many transliterations: Chebyshev, Tchebychev, Tschebyscheff, and more.

The mathematician's first name is less ambiguous: Pafnuty.

Recurrence and first few Chebyshev polynomials

To compute a general T_n it is easier to use a recurrence. Adding the two identities

$$\cos(n+1)\theta = \cos n\theta \cos \theta - \sin n\theta \sin \theta,$$

$$\cos(n-1)\theta = \cos n\theta \cos \theta + \sin n\theta \sin \theta,$$

we get $\cos(n+1)\theta + \cos(n-1)\theta = 2 \cos n\theta \cos \theta$, and therefore $T_{n+1}(x) + T_{n-1}(x) = 2x T_n(x)$. So

$$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x). \quad (7.5.3)$$

Recurrence and first few Chebyshev polynomials (cont.)

For example,

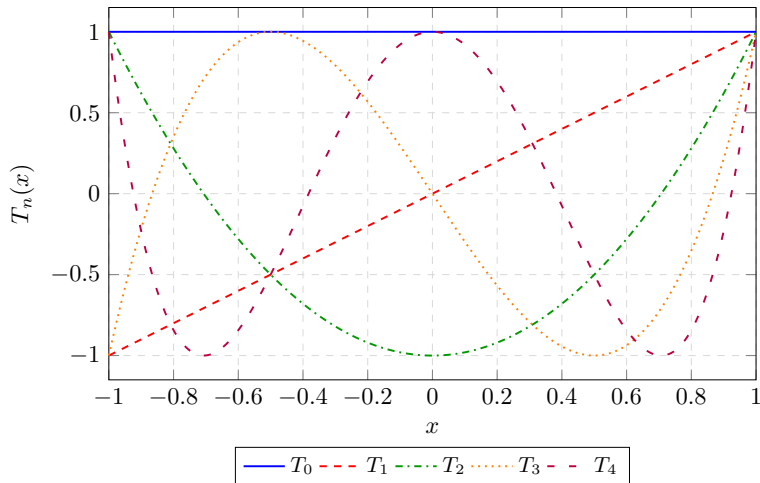
$$T_4(x) = 2x T_3 - T_2 = 2x(4x^3 - 3x) - (2x^2 - 1) = 8x^4 - 6x^2 - 2x^2 + 1 = 8x^4 - 8x^2 + 1.$$

Using this recursion we also note that $T_n(x)$ has leading coefficient 2^{n-1} for $n \geq 1$, while $T_0 = 1$. At the endpoints

$$T_n(1) = \cos(n \cdot 0) = 1, \quad T_n(-1) = \cos(n\pi) = (-1)^n,$$

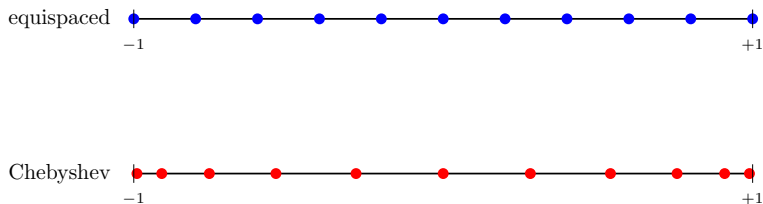
since $1 = \cos 0$ and $-1 = \cos \pi$ under $x = \cos \theta$.

Shapes of the first Chebyshev polynomials



The identity $T_n(\cos \theta) = \cos(n\theta)$ keeps every T_n between -1 and 1 on the interval $[-1, 1]$, with alternating extrema inherited from the cosine.

Chebyshev nodes



The roots $x_k = \cos((2k - 1)\pi/(2n))$, $k = 1, \dots, n$, cluster near the endpoints. This nonuniform spacing is one reason Chebyshev interpolation behaves much better near ± 1 than equally spaced interpolation.

Differential equation and orthogonality

Because $T_n(\cos \theta) = \cos n\theta$ satisfies $\frac{d^2}{d\theta^2} T_n + n^2 T_n = 0$, we convert to x using $x = \cos \theta$, so $\frac{d}{d\theta} = -\sin \theta \frac{d}{dx}$. Applying this operator twice, the outer $\frac{d}{d\theta}$ acts both on the coefficient $-\sin \theta$ (giving $\frac{d}{d\theta}(-\sin \theta) = -\cos \theta$) and on the inner $\frac{d}{dx}$ (via $\frac{d}{d\theta} = -\sin \theta \frac{d}{dx}$ again):

$$\frac{d^2}{d\theta^2} = \frac{d}{d\theta} \left(-\sin \theta \frac{d}{dx} \right) = -\cos \theta \frac{d}{dx} + (-\sin \theta)(-\sin \theta) \frac{d^2}{dx^2} = -\cos \theta \frac{d}{dx} + \sin^2 \theta \frac{d^2}{dx^2}.$$

With $\cos \theta = x$ and $\sin^2 \theta = 1 - x^2$ this is

$$\frac{d^2}{d\theta^2} = -x \frac{d}{dx} + (1 - x^2) \frac{d^2}{dx^2}.$$

So the Chebyshev polynomials $T_n(x)$ solve

$$(1 - x^2) T_n'' - x T_n' + n^2 T_n = 0. \quad (7.5.4)$$

Differential equation and orthogonality (cont.)

Multiplying by the integrating factor $(1 - x^2)^{-1/2}$ puts this in Sturm–Liouville form,

$$\frac{d}{dx} \left(\sqrt{1 - x^2} \frac{du}{dx} \right) + \frac{n^2}{\sqrt{1 - x^2}} u = 0,$$

so the Chebyshev polynomials are orthogonal with respect to

$$(u, v) = \int_{-1}^1 \frac{uv}{\sqrt{1 - x^2}} dx.$$

Because $x = \cos \theta$ (so $dx = -\sin \theta d\theta$ and $\sqrt{1 - x^2} = \sin \theta$),

$$(T_n, T_m) = \int_0^\pi \cos n\theta \cos m\theta d\theta.$$

This is 0 for $n \neq m$. For equal indices, the norm is π when $n = m = 0$ and $\pi/2$ when $n = m \geq 1$.

Useful formulas for the Chebyshev polynomial

differential equation $(1 - x^2) T_n'' - x T_n' + n^2 T_n = 0$

series solution ($n \geq 1$)
$$T_n(x) = \frac{n}{2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (n-k-1)!}{k! (n-2k)!} (2x)^{n-2k}$$

generating function
$$\frac{1 - t^2}{1 - 2xt + t^2} = T_0(x) + 2 \sum_{n=1}^{\infty} T_n(x) t^n, \quad |t| < 1$$

recurrence
$$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x)$$

orthogonality
$$\int_{-1}^1 T_n T_m (1 - x^2)^{-1/2} dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases}$$

(The generating function for T_n is $\frac{1 - xt}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} T_n(x) t^n$ for $|t| < 1$; the symmetric form above weights $n = 0$ separately.)

Summary: four orthogonal families

Family	Interval	Weight $w(x)$	Arises in
Legendre P_n	$[-1, 1]$	1	spherical harmonics
Hermite H_n	$(-\infty, \infty)$	e^{-x^2}	quantum oscillator
Laguerre L_n	$[0, \infty)$	e^{-x}	hydrogen radial part
Chebyshev T_n	$[-1, 1]$	$(1 - x^2)^{-1/2}$	$\cos n\theta$, interpolation

The shared structure is a second-order Sturm–Liouville ODE, a generating function, a recurrence, and an orthogonality relation on the natural interval and weight. For Legendre, Hermite, and Laguerre we also recorded Rodrigues' formulas and Schläfli contour integrals. Completeness then lets us expand reasonable functions in the corresponding series, exactly as Fourier series expand in e^{inx} .